

第四章 随机变量的数字特征

§ 1 随机变量的期望

§ 2 方差和标准差

§ 3 协方差和相关系数

§ 4 条件期望和预测



讨论 r.v X, Y 之间的关系

设 $(X, Y) \sim f(x, y), X \sim f_X(x), Y \sim f_Y(y)$, 则

$$X, Y \text{ 相互独立} \iff \underline{f(x, y) = f_X(x) f_Y(y)}$$

 问 若 X, Y 不独立, 如何刻画它们之间的关系 ?

分析 若 X, Y 相互独立, 则

$$E[(X - E(X))(Y - E(Y))] = 0$$

故若

$$\underline{E[(X - E(X))(Y - E(Y))] \neq 0}$$

则 X, Y 必不独立, 它们之间必存在一定关系.

定义 若 X, Y 的方差均存在, 记

$$\text{Cov}(X, Y) \triangleq E[(X - E(X))(Y - E(Y))]$$

称 $\text{Cov}(X, Y)$ 为 X, Y 的**协方差**.

协方差的基本性质

$$\text{Cov}(X, Y) = E[(X - E(X))(Y - E(Y))]$$

1 若 X, Y 相互独立 $\Rightarrow \text{Cov}(X, Y) = 0$

2 $\text{Cov}(X, Y) = \text{Cov}(Y, X)$

3 $D(X) = E(X - E(X))^2 = \text{Cov}(X, X)$

4
$$\begin{aligned}\text{Cov}(X, Y) &= E[(X - E(X))(Y - E(Y))] \\ &= E[XY - XE(Y) - YE(X) + E(X)E(Y)] \\ &= E(XY) - E[XE(Y)] - E[YE(X)] + E(X)E(Y) \\ &= E(XY) - E(X)E(Y)\end{aligned}$$

协方差的计算

$$\text{Cov}(X, Y) = E[(X - E(X))(Y - E(Y))]$$

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y)$$

5
$$\begin{aligned}D(X + Y) &= D(X) + D(Y) + 2E[(X - E(X))(Y - E(Y))] \\ &= D(X) + D(Y) + 2\text{Cov}(X, Y)\end{aligned}$$

协方差的基本性质

$$\text{Cov}(X, Y) = E[(X - E(X))(Y - E(Y))]$$

6 对任意常数 a, b 有

$$\begin{aligned}\text{Cov}(aX, bY) &= E[(aX - E(aX))(bY - E(bY))] \\ &= abE[(X - E(X))(Y - E(Y))] \\ &= ab\text{Cov}(X, Y)\end{aligned}$$

7 $\text{Cov}(X_1 + X_2, Y) = \text{Cov}(X_1, Y) + \text{Cov}(X_2, Y)$

定理 (协方差的双线性性, bilinear property)

假设 $U = a + \sum_{i=1}^n b_i X_i$ 和 $V = c + \sum_{j=1}^m d_j Y_j$, 那么

$$\text{Cov}(U, V) = \sum_{i=1}^n \sum_{j=1}^m b_i d_j \text{Cov}(X_i, Y_j).$$

协方差的意义

$\because X, Y$ 相互独立 $\Rightarrow \text{Cov}(X, Y) = 0$

$\therefore \text{Cov}(X, Y) \neq 0 \Rightarrow X, Y$ 必不独立

$\Rightarrow X, Y$ 之间必存在某种 “关系”

又 $\forall k \in R$

$$\text{Cov}(kX, kY) = k^2 \text{Cov}(X, Y)$$

考虑 “单位化” 的 r.v

$$X^* \triangleq \frac{X - E(X)}{\sqrt{D(X)}}, \quad Y^* \triangleq \frac{Y - E(Y)}{\sqrt{D(Y)}}$$

定义 称

$$\rho_{XY} \triangleq \text{Cov}(X^*, Y^*) = \frac{\text{Cov}(X, Y)}{\sqrt{D(X)}\sqrt{D(Y)}}$$

为 X, Y 的 **相关系数**.

 **问** 相关系数 ρ_{XY} 刻画了 X, Y 之间什么关系 ?

分析 考虑 X, Y 之间的线性关系, 即用 r.v

$$\hat{Y} = a + bX \quad (a, b \text{ 为常数})$$

近似地表示 r.v Y . 记 **均方误差**

$$\begin{aligned} e &= E[(Y - \hat{Y})^2] = E[(Y - (a + bX))^2] \\ &= E(Y^2) + b^2 E(X^2) + a^2 - 2bE(XY) + 2abE(X) - 2aE(Y) \end{aligned}$$

令
$$\begin{cases} \frac{\partial e}{\partial a} = 2a + 2bE(X) - 2E(Y) = 0 \\ \frac{\partial e}{\partial b} = 2bE(X^2) - 2E(XY) + 2aE(X) = 0 \end{cases}$$

解得

$$\begin{cases} b_0 = \frac{\text{Cov}(X, Y)}{D(X)} \\ a_0 = E(Y) - b_0 E(X) = E(Y) - E(X) \frac{\text{Cov}(X, Y)}{D(X)} \end{cases}$$

$$\begin{aligned}\min_{a,b} e &= \min_{a,b} E[(Y - (a + bX))^2] = E[(Y - (a_0 + b_0 X))^2] \\&= E[(Y - (E(Y) - b_0 E(X) + b_0 X))^2] \\&= E\{[(Y - E(Y)) - b_0(X - E(X))]\^2\} \\&= E[(Y - E(Y))^2] + b_0^2 E[(X - E(X))^2] \\&\quad - 2b_0 E[(Y - E(Y))(X - E(X))] \\&= D(Y) + b_0^2 D(X) - 2b_0 \text{Cov}(X, Y) \\&= D(Y) + b_0 \text{Cov}(X, Y) - 2b_0 \text{Cov}(X, Y) \\&= D(Y) - b_0 \text{Cov}(X, Y) \\&= D(Y) \left[1 - \frac{\text{Cov}^2(X, Y)}{D(X)D(Y)} \right] \\&= D(Y)(1 - \rho_{XY}^2)\end{aligned}$$

重要关系式

即有

$$\min_{a,b} E[(Y - (a + bX))^2] = E[(Y - (a_0 + b_0X))^2] \\ = D(Y)(1 - \rho_{XY}^2)$$

其中

$$\begin{cases} b_0 = \frac{\text{Cov}(X, Y)}{D(X)} \\ a_0 = E(Y) - b_0 E(X) = E(Y) - E(X) \frac{\text{Cov}(X, Y)}{D(X)} \end{cases}$$

定理 ① $|\rho_{XY}| \leq 1$

② $|\rho_{XY}| = 1 \iff Y \stackrel{a.e}{=} a + bX$ (a, b 为常数)

相关系数的实际意义

$$\because \min_{a,b} e = \min_{a,b} E[(Y - (a + bX))^2] = D(Y)(1 - \rho_{XY}^2)$$

$\therefore |\rho_{XY}|$ 较大时均方误差 e 较小



X, Y 之间的线性关系较密切

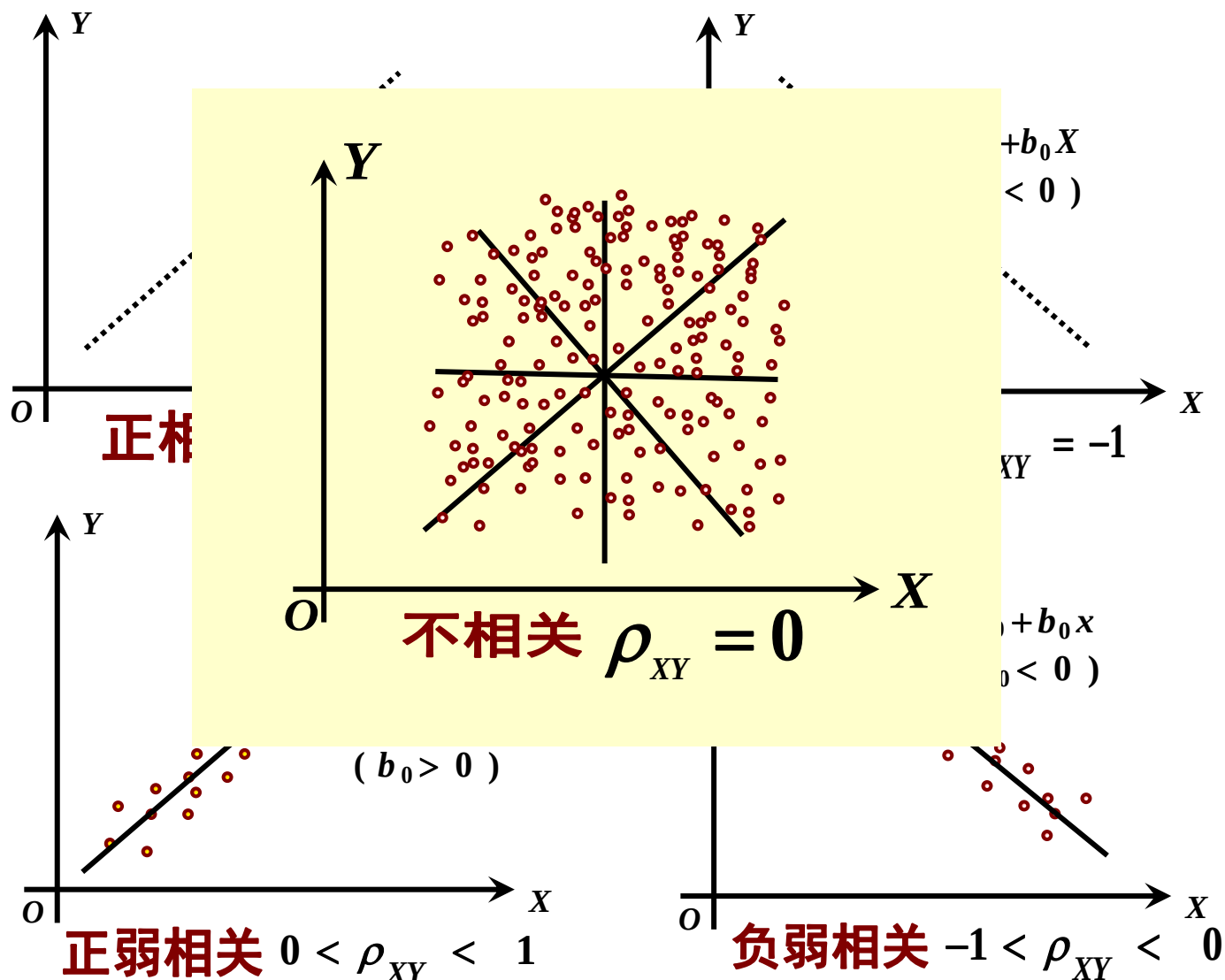
特别当 $|\rho_{XY}| = 1$ 时, X, Y 之间几乎就是线性关系

反之, 当 $|\rho_{XY}|$ 较小时, X, Y 之间的线性关系较弱

定义 当 $\rho_{XY} = 1$ 时, 称 X 与 Y **正相关**
当 $\rho_{XY} = -1$ 时, 称 X 与 Y **负相关**
当 $\rho_{XY} = 0$ 时, 称 X 与 Y **不相关**

} **相关**

r. v X Y 之间线性关系的图示



18. 设二维随机变量 (X, Y) 的密度函数为

$$f(x, y) = \begin{cases} x + y, & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0, & \text{其他} \end{cases}$$

求 $E(X)$, $E(Y)$, $E(XY)$, $\text{Cov}(X, Y)$.

$$\begin{aligned} E(X) &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} xf(x, y) dx dy \\ &= \int_0^1 \int_0^1 x(x + y) dx dy = \frac{7}{12} \end{aligned}$$

$$E(Y) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} yf(x, y) dx dy = \frac{7}{12}$$

$$\begin{aligned} E(XY) &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} xyf(x, y) dx dy \\ &= \int_0^1 \int_0^1 xy(x + y) dx dy = \frac{1}{3} \end{aligned}$$

18. 设二维随机变量 (X, Y) 的密度函数为

$$f(x, y) = \begin{cases} x + y, & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0, & \text{其他} \end{cases}$$

求 $E(X)$, $E(Y)$, $E(XY)$, $\text{Cov}(X, Y)$.

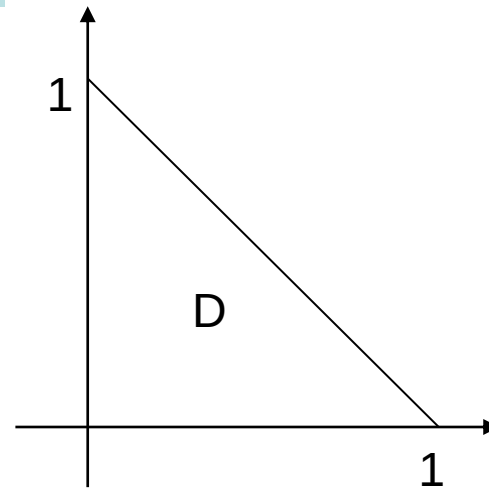
$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = -1/144$$

19. (X, Y) 在 D 上服从均匀分布,
求 $\text{cov}(X, Y), \rho_{XY}$.

解: 区域 D 的面积为 $\frac{1}{2}$

所以, (X, Y) 的联合密度为

$$f(x, y) = \begin{cases} 2 & (x, y) \in D \\ 0 & (x, y) \notin D \end{cases}$$



$$\text{cov}(X, Y) = E(XY) - E(X)E(Y) = -\frac{1}{36}.$$

$$\rho_{XY} = \frac{\text{cov}(X, Y)}{\sqrt{D(X)}\sqrt{D(Y)}} = -\frac{1}{2}$$

仿照18题

23. 设随机变量 (X, Y) 具有概率密度

$$f(x, y) = \begin{cases} \frac{1}{8}(x+y), & 0 \leq x \leq 2, 0 \leq y \leq 2 \\ 0, & \text{其他} \end{cases}$$

求 EX, EY, ρ_{XY} .

$$E(X) = \int_0^2 dx \int_0^2 x \cdot \frac{1}{8}(x+y) dy = \frac{7}{6}$$

$$E(Y) = \int_0^2 dx \int_0^2 y \cdot \frac{1}{8}(x+y) dy = \frac{7}{6}$$

$$\begin{aligned} \text{cov}(X, Y) &= E(XY) - E(X)E(Y) \\ &= \int_0^2 dx \int_0^2 xy \cdot \frac{1}{8}(x+y) dy - \frac{7}{6} \cdot \frac{7}{6} = -\frac{1}{36} \end{aligned}$$

$$\begin{aligned} D(X) &= E(X^2) - [E(X)]^2 \\ &= \int_0^2 dx \int_0^2 x^2 \cdot \frac{1}{8} (x+y) dy - \left(\frac{7}{6}\right)^2 = \frac{11}{36} \end{aligned}$$

$$D(Y) = \frac{11}{36}$$

$$\rho_{XY} = \frac{\text{cov}(X, Y)}{\sqrt{DX} \sqrt{DY}} = \frac{-\frac{1}{36}}{\frac{11}{36}} = -\frac{1}{11}$$

例 设 r.v. (X, Y) 服从圆域 $G: x^2 + y^2 \leq 1$ 上的均匀分布, 试讨论 X, Y 的独立性与相关性.

解 先求得 X, Y 的密度函数分别为

$$f_X(x) = \begin{cases} \frac{2}{\pi} \sqrt{1-x^2}, & |x| < 1 \\ 0, & |x| \geq 1 \end{cases} \quad f_Y(y) = \begin{cases} \frac{2}{\pi} \sqrt{1-y^2}, & |y| < 1 \\ 0, & |y| \geq 1 \end{cases}$$

$$\because f(x, y) \neq f_X(x) f_Y(y) \quad (\forall |x| < 1, |y| < 1)$$

$\therefore X, Y$ 不独立.

又因为

$$E(X) = \int_{-\infty}^{\infty} x f_X(x) dx = 0$$

$$E(Y) = \int_{-\infty}^{\infty} y f_Y(y) dy = 0$$

$$E(XY) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f(x, y) dx dy = \frac{1}{\pi} \iint_{x^2+y^2 \leq 1} xy dx dy = 0$$

$$\therefore E[(X - E(X))(Y - E(Y))] = E(XY) - E(X)E(Y) = 0$$

故 X, Y 不相关.

设 X, Y 相互独立, 且都服从 $N(\mu, \sigma^2)$,
 $U = aX + bY$, $V = aX - bY$,
 a, b 为常数, 且都不为零, 求 ρ_{UV}

解
$$\begin{aligned}\text{cov}(U, V) &= E(UV) - E(U)E(V) \\ &= a^2 E(X^2) - b^2 E(Y^2) \\ &\quad - [aE(X) + bE(Y)][aE(X) - bE(Y)]\end{aligned}$$

$$\text{由 } \left. \begin{array}{l} E(X) = E(Y) = \mu, \\ D(X) = D(Y) = \sigma^2 \end{array} \right\} \longrightarrow \begin{array}{l} E(X^2) = \sigma^2 + \mu^2 \\ E(Y^2) = \sigma^2 + \mu^2 \end{array}$$

$$\longrightarrow \text{cov}(U, V) = (a^2 - b^2)\sigma^2$$

$$\text{而 } D(U) = a^2 D(X) + b^2 D(Y) = (a^2 + b^2) \sigma^2$$

$$D(V) = a^2 D(X) + b^2 D(Y) = (a^2 + b^2) \sigma^2$$

$$\rho_{UV} = \frac{\text{cov}(U, V)}{\sqrt{D(U)} \sqrt{D(V)}}$$

$$\text{故 } \rho_{UV} = \frac{a^2 - b^2}{a^2 + b^2}$$

思考：还有其它方法吗？

利用协方差的性质

$$\begin{aligned} & \text{cov}(aX + bY, aX - bY) \\ &= a^2 \text{cov}(X, X) - b^2 \text{cov}(Y, Y) \\ &= a^2 D(X) - b^2 D(Y) \end{aligned}$$

$$\longrightarrow \text{cov}(U, V) = (a^2 - b^2)\sigma^2$$

例 设 $(X, Y) \sim N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$, 求 X, Y 的相关系数.

解 $\text{Cov}(X, Y) = E[(X - \mu_1)(Y - \mu_2)]$

$$\begin{aligned} \text{1} \quad f(x, y) &= \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}\right. \\ &\quad \times \left[\frac{(x-\mu_1)^2}{\sigma_1^2} - 2\rho\frac{(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2} + \frac{(y-\mu_2)^2}{\sigma_2^2}\right]\bigg\} \end{aligned}$$

$$\text{2} \quad X \sim N(\mu_1, \sigma_1^2), Y \sim N(\mu_2, \sigma_2^2)$$

$$\text{3} \quad E(X) = \mu_1, E(Y) = \mu_2$$

$$D(X) = \sigma_1^2, D(Y) = \sigma_2^2$$

猜想 $\rho = \rho_{XY}$

例 设 $(X, Y) \sim N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$, 求 X, Y 的相关系数.

解 $\text{Cov}(X, Y) = E[(X - \mu_1)(Y - \mu_2)]$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_1)(y - \mu_2) f(x, y) dx dy$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (\sigma_1 \sigma_2 \sqrt{1 - \rho^2} tu + \rho \sigma_1 \sigma_2 u^2) e^{-(t^2 + u^2)/2} dt du$$

$$f(x, y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}\left[\frac{(x-\mu_1)^2}{\sigma_1^2} - 2\rho\frac{(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2} + \frac{(y-\mu_2)^2}{\sigma_2^2}\right]\right\}$$

$$= \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}\left[\left(\frac{y-\mu_2}{\sigma_2} - \rho\frac{x-\mu_1}{\sigma_1}\right)^2 + (1-\rho^2)\frac{(x-\mu_1)^2}{\sigma_1^2}\right]\right\}$$

令 $t = \frac{1}{\sqrt{1-\rho^2}}\left(\frac{y-\mu_2}{\sigma_2} - \rho\frac{x-\mu_1}{\sigma_1}\right)$, $u = \frac{x-\mu_1}{\sigma_1}$, $J = 1$

例 设 $(X, Y) \sim N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$, 求 X, Y 的相关系数.

解

$$\begin{aligned}\text{Cov}(X, Y) &= E[(X - \mu_1)(Y - \mu_2)] \\&= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_1)(y - \mu_2) f(x, y) dx dy \\&= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (\sigma_1 \sigma_2 \sqrt{1 - \rho^2} tu + \rho \sigma_1 \sigma_2 u^2) e^{-(t^2 + u^2)/2} dt du \\&= \frac{\rho \sigma_1 \sigma_2}{2\pi} \int_{-\infty}^{\infty} e^{-t^2/2} dt \cdot \int_{-\infty}^{\infty} u^2 e^{-u^2/2} du \\&\quad + \frac{\sigma_1 \sigma_2 \sqrt{1 - \rho^2}}{2\pi} \int_{-\infty}^{\infty} t e^{-t^2/2} dt \cdot \int_{-\infty}^{\infty} u e^{-u^2/2} du \\&= \frac{\rho \sigma_1 \sigma_2}{2\pi} \sqrt{2\pi} \cdot \sqrt{2\pi} = \rho \sigma_1 \sigma_2\end{aligned}$$

$$\therefore \rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{D(X)}\sqrt{D(Y)}} = \frac{\rho \sigma_1 \sigma_2}{\sigma_1 \sigma_2} = \rho$$

$$22. Y = aX + b, (a > 0) \Rightarrow \rho_{XY} = 1$$

$$\begin{aligned}\operatorname{cov}(X, Y) &= \operatorname{cov}(X, aX + b) \\ &= a \operatorname{cov}(X, X) = aD(X)\end{aligned}$$

$$D(Y) = a^2 D(X)$$

$$\rho_{XY} = \frac{\operatorname{cov}(X, Y)}{\sqrt{D(X)} \sqrt{D(Y)}} = \frac{aD(X)}{aD(X)} = 1$$

独立与不相关的关系

X, Y 相互独立  X, Y 互不相关

指 X, Y 之间没有任何关系,当然也没有线性关系

指 X, Y 之间没有线性关系,但可能有其它关系

特例 设 $(X, Y) \sim N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$, 则

X, Y 相互独立  $\rho = 0$

 $\rho_{XY} = 0$

 X, Y 互不相关

定义 对于r.v X, Y , 称

$$E(X^k) \quad (k = 1, 2, \dots)$$

为 **k 阶原点矩**, 简称 **k 阶矩**. 称

$$E[(X - E(X))^k] \quad (k = 2, 3, \dots)$$

为 **k 阶中心矩**. 称

$$E(X^k Y^l) \quad (k, l = 1, 2, \dots)$$

为 **$k + l$ 阶混合矩**. 称

$$E[(X - E(X))^k (Y - E(Y))^l] \quad (k, l = 1, 2, \dots)$$

为 **$k + l$ 阶混合中心矩**.

$E(X)$ 1 阶原点矩

$D(X)$ 2 阶中心矩

$\text{Cov}(X, Y)$ 2 阶混合中心矩

对于二维r.v. (X_1, X_2) , 记

$$c_{11} = E[(X_1 - E(X_1))^2] = D(X_1)$$

$$c_{12} = E[(X_1 - E(X_1))(X_2 - E(X_2))] = \text{Cov}(X_1, X_2)$$

$$c_{21} = E[(X_2 - E(X_2))(X_1 - E(X_1))] = \text{Cov}(X_2, X_1)$$

$$c_{22} = E[(X_2 - E(X_2))^2] = D(X_2)$$

写成矩阵的形式

$$C \triangleq \text{Cov}(X, X) \triangleq \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix}$$

称矩阵 $\text{Cov}(X, X)$ 为 $X \triangleq (X_1, X_2)$ 的**协方差矩阵**.

易知 ① $C^T = C$, 即 C 为对称阵

② $C \geq 0$, 即 C 为正定(非负定)阵

证 $\Delta_1 = c_{11} = D(X_1) \geq 0$

$$\Delta_2 = |C| = c_{11}c_{22} - c_{21}c_{12}$$

$$= D(X_1)D(X_2) - [\text{Cov}(X_1, X_2)]^2$$

$$= D(X_1)D(X_2)(1 - \rho_{X_1X_2}^2) \geq 0$$

总结：协方差和相关系数的性质

$$(1) \quad \text{cov}(X, Y) = \text{cov}(Y, X)$$

$$(2) \quad \text{cov}(aX, bY) = ab \text{cov}(X, Y)$$

$$(3) \quad \text{cov}(X + Y, Z) = \text{cov}(X, Z) + \text{cov}(Y, Z)$$

$$(4) \quad \text{cov}(X, X) = D(X)$$

$$(5) \quad D(X \pm Y) = D(X) + D(Y) \pm 2 \text{cov}(X, Y)$$

$$(6) \quad |\rho_{XY}| \leq 1$$

$$(7) \quad |\rho_{XY}| = 1 \iff \text{存在常数 } a, b (a \neq 0), \\ \text{使 } P(Y = aX + b) = 1,$$

即 X 和 Y 以概率 1 线性相关.

X, Y 相互独立 $\not\iff X, Y$ 不相关

因为 $\rho_{XY} = 0 \iff X, Y$ 不相关

$\iff \text{cov}(X, Y) = 0$

若 (X, Y) 服从二维正态分布,

X, Y 相互独立 $\iff X, Y$ 不相关



P119: 54、60、补充题

设随机变量 X 的概率密度为 $f(x) = \frac{1}{2}e^{-|x|}, -\infty < x < +\infty$,

(1) 求出 $E(X), D(X)$.

(2) X 与 $|X|$ 是否独立? 说明理由.

(3) X 与 $|X|$ 是否相关? 说明理由.

1. 设随机变量 (X, Y) 的概率密度为

$$f(x, y) = \begin{cases} \frac{1}{8}(x + y), & 0 \leq x \leq 2, 0 \leq y \leq 2, \\ 0, & \text{其他.} \end{cases}$$

求 $E(X), E(Y), Cov(X, Y), \rho_{XY}, D(X + Y)$.

2. 设随机变量 X 和 Y 独立同分布于 $N(\mu, \sigma^2)$. 令 $Z = \alpha X + \beta Y$,
 $W = \alpha X - \beta Y$, 求 $Cov(Z, W), \rho_{ZW}$.